AdaBoost - Complete Worked Example

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Abstract

This document walks through a full, step-by-step AdaBoost example (binary classification) using four simple decision stumps (CGPA, Interactiveness, Practical, Communication). Every numeric step is shown: initial weights, weighted errors, classifier weights α_t, unnormalized and normalized weight updates, and the final combined classifier evaluation on each training example. Explanations are provided in plain language at each stage so you can follow the logic.

1 Problem and dataset

We consider a binary classification problem (target: job offer 'yes' or 'no') and encode:

$$\text{yes} = +1, \qquad \text{no} = -1.$$

The training set ($N = 6$) is:

Table 1: Training data

i	CGPA ≥ 9?	Interactiveness	Practical	Communication	y_i
1	yes	yes	good	good	+1
2	no	no	good	moderate	+1
3	yes	no	average	moderate	-1
4	no	no	average	good	-1
5	yes	yes	good	moderate	+1
6	yes	yes	good	moderate	+1

All samples start with equal weight:

$$w_i^{(1)} = \frac{1}{N} = \frac{1}{6} \approx 0.1666667, \qquad i = 1, \ldots, 6.$$

Below we run AdaBoost rounds. For clarity we show for each example:

- the weak stump prediction $h_t(x_i)$,
- the product $y_i h_t(x_i)$ (which is $+1$ when correctly classified, -1 when misclassified),
- how the weight update multiplies each sample by either $e^{-\alpha_t}$ (correct) or $e^{+\alpha_t}$ (wrong),
- the normalization constant Z_t and normalized weights $w_i^{(t+1)}$.

2 Round 1: stump h_1 (CGPA $\geq 9 \Rightarrow +1$, else -1)

Stump definition and predictions. We fix the first stump as:

$$h_1(x) = \begin{cases} +1 & \text{if CGPA} \geq 9, \\ -1 & \text{otherwise.} \end{cases}$$

Predictions for the 6 examples:

$$h_1(x) = [+1,\ -1,\ +1,\ -1,\ +1,\ +1].$$

Per-example comparison and sign values. We compute $y_i h_1(x_i)$ (this is $+1$ if the stump is correct for example i, -1 if wrong):

i	y_i	$h_1(x_i)$	$y_i h_1(x_i)$	initial $w_i^{(1)}$
1	+1	+1	+1	0.1666667
2	+1	-1	-1	0.1666667
3	-1	+1	-1	0.1666667
4	-1	-1	+1	0.1666667
5	+1	+1	+1	0.1666667
6	+1	+1	+1	0.1666667

Misclassified indices: $i = 2, 3$.

Weighted error and classifier weight. Weighted error:

$$\epsilon_1 = \sum_{i:y_i \neq h_1(x_i)} w_i^{(1)} = w_2^{(1)} + w_3^{(1)} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \approx 0.3333333.$$

Classifier weight:

$$\alpha_1 = \frac{1}{2} \ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right) = \frac{1}{2} \ln\left(\frac{2/3}{1/3}\right) = \frac{1}{2} \ln 2 \approx 0.3465736.$$

Interpretation: $\alpha_1 > 0$ means h_1 is better than random (required for AdaBoost). Larger α means more influence in the final vote.

Update factors. We use:

$e^{-\alpha_1} \approx 0.7071068$ (applied when $y_i h_1(x_i) = +1$), $\quad e^{+\alpha_1} \approx 1.4142136$ (applied when $y_i h_1(x_i) = -1$

Unnormalized updated weights (per example).

$$\tilde{w}_i = w_i^{(1)} e^{-\alpha_1 y_i h_1(x_i)}.$$

Computing:

$\tilde{w} = [0.1666667 \cdot 0.7071068,\ 0.1666667 \cdot 1.4142136,\ 0.1666667 \cdot 1.4142136,\ 0.1666667 \cdot 0.7071068,\ 0.166666$

Numerically:

$$\tilde{w} \approx [0.1178511,\ 0.2357023,\ 0.2357023,\ 0.1178511,\ 0.1178511,\ 0.1178511].$$

Normalization.

$$Z_1 = \sum_{i=1}^{6} \tilde{w}_i = 0.9428090416.$$

So normalized weights for next round:

$$w_i^{(2)} = \frac{\tilde{w}_i}{Z_1} \approx [\,0.125,\ 0.25,\ 0.25,\ 0.125,\ 0.125,\ 0.125\,].$$

Explanation: examples 2 and 3 increased weight (from 0.1667 to 0.25) because they were misclassified; AdaBoost will focus more on them in the next round.

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3 Round 2: stump h_2 (Interactiveness = yes $\Rightarrow +1$, else -1)

Stump and predictions.

$$h_2(x) = \begin{cases} +1 & \text{if Interactiveness} = \text{yes} \\ -1 & \text{otherwise} \end{cases}$$

Predictions:

$$h_2(x) = [+1,\ -1,\ -1,\ -1,\ +1,\ +1].$$

Per-example signs and weights entering round 2. Use $w^{(2)}$ from previous round:

$$w^{(2)} = [0.125,\ 0.25,\ 0.25,\ 0.125,\ 0.125,\ 0.125].$$

Compute $y_i h_2(x_i)$:

i	y_i	$h_2(x_i)$	$y_i h_2(x_i)$	$w_i^{(2)}$
1	+1	+1	+1	0.125
2	+1	-1	-1	0.25
3	-1	-1	+1	0.25
4	-1	-1	+1	0.125
5	+1	+1	+1	0.125
6	+1	+1	+1	0.125

Misclassified: $i = 2$ only (as you stated).

Weighted error and α_2.

$$\epsilon_2 = w_2^{(2)} = 0.25.$$

$$\alpha_2 = \frac{1}{2} \ln\left(\frac{1-\epsilon_2}{\epsilon_2}\right) = \frac{1}{2}\ln(3) \approx 0.5493061.$$

Interpretation: h_2 is reasonably strong (better than h_1 here) and will get a larger vote weight.

Update multipliers.

$$e^{-\alpha_2} \approx 0.5773503, \qquad e^{+\alpha_2} \approx 1.7320508.$$

Unnormalized weights after round 2 (per example).

$$\tilde{w}_i = w_i^{(2)} e^{-\alpha_2 y_i h_2(x_i)}.$$

Compute numerically:

$$\tilde{w} \approx [\,0.0721688,\ 0.4330127,\ 0.1443376,\ 0.0721688,\ 0.0721688,\ 0.0721688\,].$$

Normalization.

$$Z_2 = \sum_i \tilde{w}_i \approx 0.8660254038.$$

Normalized weights:

$$w^{(3)} = \frac{\tilde{w}}{Z_2} \approx [\,0.0833333,\ 0.5,\ 0.1666667,\ 0.0833333,\ 0.0833333,\ 0.0833333\,].$$

Explanation: example 2 is now the heaviest (0.5), meaning the next stump will likely try to correct it.

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4 Round 3: stump h_3 (Practical = good $\Rightarrow +1$, else -1)

Stump and predictions.

$$h_3(x) = \begin{cases} +1 & \text{if Practical} = \text{good} \\ -1 & \text{if Practical} = \text{average} \end{cases}$$

Predictions:

$$h_3(x) = [+1,\ +1,\ -1,\ -1,\ +1,\ +1].$$

Per-example sign check with weights entering round 3. Weights before round 3:

$$w^{(3)} \approx [0.0833333,\ 0.5,\ 0.1666667,\ 0.0833333,\ 0.0833333,\ 0.0833333].$$

Compute $y_i h_3(x_i)$:

i	y_i	$h_3(x_i)$	$y_i h_3(x_i)$	$w_i^{(3)}$
1	+1	+1	+1	0.0833333
2	+1	+1	+1	0.5
3	-1	-1	+1	0.1666667
4	-1	-1	+1	0.0833333
5	+1	+1	+1	0.0833333
6	+1	+1	+1	0.0833333

All entries $y_i h_3(x_i) = +1 \Rightarrow h_3$ classifies every training example correctly.

Weighted error.

$$\epsilon_3 = \sum_{i:y_i \neq h_3(x_i)} w_i^{(3)} = 0.$$

Classifier weight. Formally,

$$\alpha_3 = \frac{1}{2} \ln\left(\frac{1-\epsilon_3}{\epsilon_3}\right) = \frac{1}{2} \ln\left(\frac{1}{0}\right) \to +\infty.$$

Meaning: h_3 is a perfect classifier on the training set. In practice AdaBoost stops here because we have zero training error; the final classifier will correctly label all training samples.

Practical continuation (if forced). If you must continue to show numeric values (for demonstration), add a tiny smoothing $\delta = 10^{-12}$ and treat $\epsilon_3 = \delta$, giving a finite but very large

$$\alpha_3 \approx \frac{1}{2} \ln\left(\frac{1-\delta}{\delta}\right) \approx 13.8155.$$

Because all $y_i h_3(x_i) = +1$, the unnormalized update multiplies every weight by the same factor $e^{-\alpha_3}$, and so after normalization the relative weights remain identical to $w^{(3)}$.

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5 (Optional) Round 4: stump h_4 (Communication)

Choice and polarity. We search polarity that minimizes weighted error on $w^{(3)}$. The best polarity here is:

$$\text{if Communication} = \text{good then } h_4(x) = -1, \text{ else } +1.$$

(That choice gives the smallest weighted error under $w^{(3)}$.)

Predictions:

$$h_4(x) = [-1,\ +1,\ +1,\ -1,\ +1,\ +1].$$

Per-example signs (using $w^{(3)}$):

i	y_i	$h_4(x_i)$	$y_i h_4(x_i)$	$w_i^{(3)}$
1	+1	-1	-1	0.0833333
2	+1	+1	+1	0.5
3	-1	+1	-1	0.1666667
4	-1	-1	+1	0.0833333
5	+1	+1	+1	0.0833333
6	+1	+1	+1	0.0833333

Misclassified: $i = 1$ and $i = 3$.

Weighted error and α_4.

$$\epsilon_4 = w_1^{(3)} + w_3^{(3)} = 0.0833333 + 0.1666667 = 0.25.$$

$$\alpha_4 = \frac{1}{2}\ln\left(\frac{1-0.25}{0.25}\right) = \frac{1}{2}\ln 3 \approx 0.5493061.$$

Unnormalized and normalized updates follow the same pattern as prior rounds (we computed them earlier in the numeric walk-through).

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6 Final strong classifier and per-example final decision (combined rule)

Combined classifier formula. After T rounds (here up to 4 for demonstration), AdaBoost produces the strong classifier:

$$H(x) = \text{sign}\Big(f(x)\Big), \qquad f(x) = \sum_{t=1}^{T} \alpha_t\, h_t(x).$$

Interpretation:

- Each weak learner h_t predicts either $+1$ or -1.
- Each α_t is the weight (confidence) of h_t. The sign of $f(x)$ decides the final label.
- If $f(x) > 0$ we predict $+1$ ("yes"), if $f(x) < 0$ we predict -1 ("no"). If $f(x) = 0$ the classifier is exactly undecided (rare); tie-breaking by convention can go to $+1$ or -1.
- The magnitude $|f(x)|$ indicates confidence: larger magnitude $\Rightarrow$ stronger consensus among weighted weak learners.

Values of α_t used (numeric). From the rounds above we have:

$$\alpha_1 \approx 0.3465736, \quad \alpha_2 \approx 0.5493061, \quad \alpha_3 \approx 13.8155106 \text{ (if smoothed)}, \quad \alpha_4 \approx 0.5493061.$$

Note: α_3 is huge because h_3 was perfect; in real AdaBoost we stop and set $H(x) = h_3(x)$.

Per-example computation of $f(x)$ (forced 4-round ensemble for demonstration). We compute contributions $\alpha_t h_t(x_i)$ and sum them.

i	$h_1(x_i)$	$h_2(x_i)$	$h_3(x_i)$	$h_4(x_i)$	$f(x_i) = \sum \alpha_t h_t(x_i)$	$H(x_i) = \text{sign}$
1	+1	+1	+1	-1	$0.3466 + 0.5493 + 13.8155 - 0.5493 \approx 14.1621$	+1
2	-1	-1	+1	+1	$-0.3466 - 0.5493 + 13.8155 + 0.5493 \approx 13.4689$	+1
3	+1	-1	-1	+1	$0.3466 - 0.5493 - 13.8155 + 0.5493 \approx -13.4689$	-1
4	-1	-1	-1	-1	$-0.3466 - 0.5493 - 13.8155 - 0.5493 \approx -15.2607$	-1
5	+1	+1	+1	+1	$0.3466 + 0.5493 + 13.8155 + 0.5493 \approx 15.2607$	+1
6	+1	+1	+1	+1	$0.3466 + 0.5493 + 13.8155 + 0.5493 \approx 15.2607$	+1

Explanation of the final decision rule (textual, logical).

- For each example i, every stump h_t casts a "vote" of $+1$ or -1. We multiply each vote by its reliability α_t and sum them to obtain $f(x_i)$.
- If $f(x_i) > 0$, the weighted votes favor class $+1$ sufficiently and we predict $+1$; if $f(x_i) < 0$ we predict -1.
- Example (i=3): $f(x_3) \approx -13.4689 < 0$, so the ensemble predicts -1 ("no"). That matches the true label $y_3 = -1$.
- Note the huge α_3: because h_3 perfectly matches the training labels, its vote dominates the sum, so $H(x)$ essentially follows h_3 on the training set.
- The sign is literal: **negative sum $\Rightarrow$ predict negative**, **positive sum $\Rightarrow$ predict positive**.

7 What does α_t mean (intuitively)?

- α_t is the *weight* assigned to the t-th weak learner in the final ensemble. It depends on the weighted error ϵ_t via:

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right).$$

- If ϵ_t is small (the learner is good), α_t is large and that learner's predictions have greater influence.
- If $\epsilon_t = 0.5$ (random guessing), $\alpha_t = 0$ and the learner contributes nothing.
- If $\epsilon_t > 0.5$ (worse than random), α_t becomes negative (the learner is reversed to make it useful).
- The final score $f(x)$ is a weighted sum of the learners — α_t scales their votes according to reliability.

8 Practical notes and stopping condition

- When a round produces $\epsilon_t = 0$ (perfect weak learner), AdaBoost typically *stops*, because the perfect learner already classifies the training set correctly. The final classifier can simply be that perfect stump.
- In our demonstration we continued numerically using a smoothing δ so we could show the arithmetic for a 4-round ensemble — but the correct conceptual behavior is to stop at the perfect learner.
- The sign rule is what determines labels: $\mathrm{sign}(f(x))$ returns $+1$ if $f(x) > 0$, -1 if $f(x) < 0$. If $f(x) = 0$ (rare) a tie-break rule must be chosen.

9 Summary

- We initialized equal weights and iteratively trained decision stumps.
- Round 1 (CGPA) and Round 2 (Interactiveness) increased focus on hard examples by raising their weights.
- Round 3 (Practical) was perfect on the dataset, giving $\epsilon_3 = 0$; this stump alone classifies the training set correctly, so AdaBoost would stop.
- If we combine stumps with their α_t weights, the sign of the weighted sum determines the final label. Negative sum $\Rightarrow$ negative prediction, positive sum $\Rightarrow$ positive prediction.
- The magnitude of the sum is a confidence measure.